VAMSI MURAMREDDY

SUMMARY:

- As an accomplished Product Developer with expertise in LLMOps, MLOps and AI-driven solutions. Skilled in deploying, optimizing, and maintaining AI models in production, with a focus on real-time applications in content creation and healthcare. Experienced in streamlining development workflows using MLOps tools, CI/CD pipelines, and cloud infrastructure (AWS) for scalable systems. A collaborative problem solver, dedicated to continuous model optimization and operational efficiency.

CERTIFICATIONS:

- **Microsoft Certified: Microsoft Azure AI Fundamentals (AI-900)**
  Certification Id : I577-0234   Issued On : Jan 2023
- **Microsoft Certified: Microsoft Azure Data Fundamentals (DP-900)**
  Certification Id : I594-0443   Issued On : Feb 2023
- **Microsoft Certified: Microsoft Azure Fundamentals (AZ-900)**
  Certification Id : I581-4965   Issued On : Jan 2023

TECHNICAL SKILLS:

- **Programming Languages:** Python and C
- **Web Frameworks:** Django, Flask and Fast API
- **DataBases:** Vector Databases(Pine Cone, Weviate, PG Vector), MySQL and Postgres
- **Hyper Scalers:** AWS and Azure
- **ML Libraries & Packages:** Pandas, Numpy, Scikit-Learn, Matplotlib, Plotly, AWS SDK, Evidently, Pycaret, Streamlit, Random-search, Grid-search, Optuna and Hyperopt, Streamlit, Gradio
- **LLM Libraries & Frameworks:** Pytorch, Langchain, Transformers, Hugging Face, Open AI and Open-source LLMs and NLTK, Ethical AI frameworks
- **Tools:** Jenkins, Mlflow, Kube flow, Airflow, Prometheus, Grafana, Minio, Docker, Langflow, Flowise, WandB and Amazon Sagemaker

EDUCATIONAL DETAILS:

- I earned my Bachelor's degree in Electronics and Communication Engineering from “**Audisankara College of Engineering and Technology**”, achieving a commendable **8.78** *C.G.P.A.*

PROFESSIONAL EXPERIENCE:

- Working as a **Software Engineer** (***AI/ML***) in **Cloud Angles Digital Transformation Private Limited** from **July 2022** to till date.

PROJECT EXPERIENCE:

1. **mlangles | MLOps**

    - Contributed significantly to a groundbreaking product development initiative, assuming a key role in the integration of a diverse array of open-source MLOps tools. The integrated tools proved instrumental in empowering Data Scientists and ML Engineers to seamlessly establish end-to-end processes Where the CI/CD/CT/CM will take place.

    Responsibilities and Achievements:

    - Orchestrated the integration of open-source MLOps tools, enabling efficient CI/CD pipelines and precise version control for both data and models.
    - Facilitated comprehensive model training and monitoring, ensuring a dynamic and responsive data analysis environment.
    - Played a pivotal role in orchestrating the seamless deployment of models into production environments, supporting real-time predictions and batch processing.
    - Spearheaded the development of an Auto ML tool tailored specifically for supervised learning scenarios, a notable achievement within the project.
    - Managed AWS Cloud services to facilitate the integration of various MLOps tools, ensuring a robust and scalable infrastructure for the project's success.

2. **mlangles | LLMOps:**

    - Developed an advanced Language Model Management (LLMOps) system that intelligently selects the best LLM for responses using prompt templates for accuracy. Integrated with diverse document sources and RAG operations for comprehensive insights. Enabled customization of applications and bots, tailoring LLMs to specific needs, while optimizing performance through continuous fine-tuning with proprietary data.

Responsibilities and Achievements:

- Developed a robust **RAG** system that empowers users to access information effortlessly, regardless of location. Leveraged advanced algorithms to optimize query performance and ensure accurate results. Conducted thorough evaluations of the RAG model and integrated responsible AI practices to mitigate potential biases and risks.

- Extensive hands-on experience with **text-to-text, text-to-image, and speech-to-text LLMs**, designing and implementing real-time use cases in content creation and healthcare. Delivered AI-driven solutions for automated content generation, speech recognition in healthcare workflows, and enhanced visual content creation.

- Optimized open-source LLMs (e.g., Falcon, LLama2, Orca, MPT) by applying quantization techniques (**GGUF and AWQ**) and leveraging Parameter Efficient Fine-Tuning (**PEFT**) with Lora, improving resource utilization, model performance, and adaptability across tasks.

- Implemented Ethical AI frameworks to validate user inputs and ensure responsible handling of data. Monitored and assessed LLM responses to maintain accuracy, fairness, and compliance with ethical standards. Logged metrics for each query and continuously monitored system performance for optimization and improvement.

- Designed intuitive app flows, eliminating the need for users to perform operations in the UI for different tools, thereby streamlining the querying process using normal text inputs.

- Successfully deployed and integrated multiple open-source LLMs into the product ecosystem, ensuring seamless scalability, performance, and reliability in production environments.

PERSONAL DETAILS:

- **Name :** Vamsi Muramreddy
- **Date of Birth :** 09-12-2000
- **Languages Known :** English, Hindi and Telugu
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(M. Vamsi)